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VEHICLE SEAT

Abstract

A vehicle seat may include a seat frame assembly including a seat cushion frame being folded up and down, a swivel device configured to swivel the seat frame assembly, a swivel lever configured to unlock the swivel device, and a hook rotated together with the swivel lever to fix the seat cushion frame.

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Background/Summary

CROSS-REFERENCE TO RELATED APPLICATION

[0001] This application claims under 35 U.S.C. § 119(a) the benefit of priority to Korean Patent Application Nos. 10-2024-0033968 filed on Mar. 11, 2024 and 10-2024-0141496 filed on Oct. 16, 2024, the entire contents of which are incorporated herein by reference.

BACKGROUND

(a) Technical Field

[0002] The present disclosure relates to a vehicle seat including a structure for preventing a seat cushion frame from being folded down when unlocking a seat frame assembly for swiveling.

(b) Background Art

[0003] A vehicle is provided with seats for passengers to sit on. Recently, development of autonomous vehicles and purpose-built vehicles (PBVs) is ongoing, and accordingly, spaces in vehicles may be transformed into spaces where various functions are performed.

[0004] Seats in the vehicles may be moved or rotated freely, and unnecessary seats may be stowed to create a space. If a seat in a vehicle is not in use, the seat may be folded up and stowed. However, if the seat is rotated without being folded up, interference with other structures placed in the vehicle occurs due to the rotation of the seat. In addition, even if the seat is rotated in the folded-up state, as the folded-up state of the seat is released, problems, such as difficulty in rotating the seat or damage to the seat, occur.

SUMMARY OF THE DISCLOSURE

[0005] The present disclosure has been made in an effort to solve the above-described problems associated with the prior art, and it is an object of the present disclosure to provide a vehicle seat including a structure for preventing a seat cushion frame from being folded down when unlocking a seat frame assembly for swiveling.

[0006] In one aspect, the present disclosure provides a vehicle seat including a seat frame assembly including a seat cushion frame being foldable up and down, a swivel device configured to swivel the seat frame assembly, a swivel lever configured to unlock the swivel device, and a hook rotated together with the swivel lever to fix the seat cushion frame.

[0007] In a preferred embodiment, the vehicle seat may further include a fixing pin configured to protrude toward an outside of the seat cushion frame, and the hook may be engaged with the fixing pin by rotation.

[0008] In another preferred embodiment, when the seat cushion frame is in a folded-up state and the swivel device is unlocked by the swivel lever, the hook may be engaged with the fixing pin to prevent the seat cushion frame from being folded down.

[0009] In still another preferred embodiment, the vehicle seat may further include a shaft configured to connect the hook and the swivel lever.

[0010] In yet another preferred embodiment, the vehicle seat may further include an elastic body configured to connect the swivel lever and the seat cushion frame, and the elastic body may apply force to the swivel lever so that the hook is maintained in a state of being engaged with the fixing pin.

[0011] In still yet another preferred embodiment, the vehicle seat may further include an elastic body configured to connect the swivel lever and the seat cushion frame, and when the seat cushion frame is in a folded-up state and the swivel device is in a locked state in which no external force is applied to the swivel device, the elastic body may apply force to the swivel lever such that the hook moves away from the fixing pin.

[0012] In a further preferred embodiment, when the seat frame assembly is swiveled in a folded-up state of the seat cushion frame, the hook may be rotated away from the fixing pin.

[0013] In another further preferred embodiment, the vehicle seat may further include an elastic body configured to connect the swivel lever and the seat cushion frame, and in a state where the rotation of the hook away from the fixing pin is completed, the elastic body may apply force to the

swivel lever such that the hook moves away from the fixing pin.

[0014] In still another further preferred embodiment, the swivel device may include a floor plate coupled to a floor of a vehicle or a moving rail provided on the floor of the vehicle, the floor plate includes fixing grooves, and a rotating plate provided on the floor plate and coupled to the seat frame assembly to allow the seat frame assembly to be swivelable based on external force, and the swivel lever may be connected to the rotating plate of the swivel device through a wire.

[0015] In yet another further preferred embodiment, the rotating plate may include a rotating lever connected to the wire, an inner plate including locking pin receiving grooves configured to receive locking pins so that the locking pins are movable thereinto, and the locking pins moved in conjunction with the rotating lever and moved within the locking pin receiving grooves in a direction of being inserted into the fixing grooves, and when the wire is pulled by the swivel lever, the rotating lever may be rotated, and the locking pins may be separated from the fixing grooves by the rotation of the rotating lever so that the seat frame assembly becomes swivelable.

[0016] In still yet another further preferred embodiment, the fixing grooves of the floor plate are provided as concave portions on an inner surface of the floor plate, and a plurality of guides are formed to protrude toward the inner plate on the inner surface of the floor plate, and the fixing grooves and the guides may be formed at different elevations in a direction from the swivel device toward the seat frame assembly.

[0017] In a still further preferred embodiment, one end of each of the locking pins may be inserted into one of the fixing grooves, a remaining end of each of the locking pins may come in contact with one of the guides during a process of swiveling the seat frame assembly, and wherein the one end and the remaining end of each of the locking pins may be formed at different elevations in a stepwise manner.

[0018] In a yet still further preferred embodiment, the one end of each of the locking pins may be separated from the one of the fixing grooves by the rotation of the swivel lever, the remaining end of each of the locking pins may come in contact with the one of the guides during the process of swiveling the seat frame assembly, and when the remaining end of each of the locking pins comes in contact with the one of the guides, the one end of each of the locking pins may be moved externally of the inner plate so that the wire is pulled by the rotating lever.

[0019] In another further preferred embodiment, the wire is pulled by the rotating lever so that the hook moves away from the fixing pin and the seat cushion assembly can be folded down.

[0020] In still another further preferred embodiment, the seat frame assembly may further include a center frame and a back frame, the back frame and the seat cushion frame may be coupled to the center frame, and the center frame and the swivel device may be connected such that, when the swivel device is unlocked, the center frame may become swivelable.

[0021] In yet another further preferred embodiment, the vehicle seat may further include a fixing pin configured to protrude toward an outside of the seat cushion frame, and if the seat cushion frame is swiveled in a folded-up state of the seat cushion frame, the hook may be rotated from a state of being engaged with the fixing pin to a state of being spaced apart from the fixing pin.

[0022] Other aspects and preferred embodiments of the disclosure are discussed infra.

[0023] The above and other features of the disclosure are discussed infra.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

[0024] The above and other features of the present disclosure will now be described in detail with reference to certain exemplary embodiments thereof illustrated in the accompanying drawings which are given hereinbelow by way of illustration only, and thus are not limitative of the present disclosure, and wherein:

[0025] FIG. 1 is a view showing a vehicle seat according to one embodiment of the present disclosure;

[0026] FIG. 2 is an exploded view of the vehicle seat of FIG. 1;

[0027] FIG. 3 is a view showing a swivel device according to one embodiment of the present disclosure;

[0028] FIG. 4 is a view showing the swivel device located on a moving rail according to one embodiment of the present disclosure;

[0029] FIG. 5 is a view showing a fixed frame assembly according to one embodiment of the present disclosure;

[0030] FIG. 6 is an exploded view of the fixed frame assembly of FIG. 5;

[0031] FIGS. 7 and 8 are views showing a locked state of a seat cushion frame according to one embodiment of the present disclosure;

[0032] FIG. 9 is a view showing an unlocked state of the seat cushion frame according to one embodiment of the present disclosure;

[0033] FIG. 10 is a view showing one surface of the swivel device according to one embodiment of the present disclosure;

[0034] FIG. 11 is a view showing the other surface of the swivel device according to one embodiment of the present disclosure;

[0035] FIG. 12 is a view for explaining the inner surface of a floor plate according to one embodiment of the present disclosure;

[0036] FIG. 13 is a view showing a folded-up state of the seat cushion frame according to one embodiment of the present disclosure;

[0037] FIGS. 14 and 15 are views showing an unlocked state of the swivel device according to one embodiment of the present disclosure;

[0038] FIG. 16 is a view showing the seat frame assembly according to one embodiment of the present disclosure that is swiveled;

[0039] FIGS. 17 and 18 are views showing a process of swiveling the swivel device according to one embodiment of the present disclosure; and

[0040] FIGS. 19 and 20 are views showing a state in which swiveling of the swivel device according to one embodiment of the present disclosure is completed.

[0041] It should be understood that the appended drawings are not necessarily to scale, presenting a somewhat simplified representation of various preferred features illustrative of the basic principles of the disclosure. The specific design features of the present disclosure as disclosed herein, including, for example, specific dimensions, orientations, locations, and shapes, will be determined in part by the particular intended application and use environment.

[0042] In the figures, reference numbers refer to the same or equivalent parts of the present disclosure throughout the several figures of the drawing.

DETAILED DESCRIPTION

[0043] Advantages and features of the present disclosure and methods for achieving the same will become apparent from the descriptions of the embodiments hereinbelow with reference to the accompanying drawings. However, the present disclosure is not limited to the embodiments disclosed herein and may be implemented in various different forms, and these embodiments are provided to make the description of the present disclosure thorough and to fully convey the scope of the present disclosure to those skilled in the art. It is to be noted that the scope of the present disclosure is defined only by the claims. In the following description of the embodiments, the same or similar elements are denoted by the same reference numerals even though they are depicted in different drawings.

[0044] In addition, in the following description of the embodiments, terms, such as “first” and “second”, are used only to distinguish various elements from each other because the names of the elements are the same, and do not imply a sequence or order unless clearly indicated by the

context.

[0045] The detailed description is illustrative of the present disclosure. Further, the detailed description is intended to illustrate exemplary embodiments of the present disclosure, and the present disclosure may be used in various other combinations, modifications, and environments. That is, changes or modifications can be made within the scope of the disclosure disclosed in the description, a scope equivalent to the disclosed content, and/or the scope of technology or knowledge in the art. The following embodiments illustrate the best mode for implementing the technical idea of the present disclosure, and various changes required for specific application fields and uses of the present disclosure are also possible. Accordingly, the following detailed description of the present disclosure is not intended to limit the disclosure to the disclosed embodiments. In addition, the appended claims should be construed to include other embodiments.

[0046] FIG. 1 is a view showing a vehicle seat according to one embodiment of the present disclosure, and FIG. 2 is an exploded view of the vehicle seat of FIG. 1.

[0047] Referring to FIGS. 1 and 2, the vehicle seat may include a seat frame assembly **10** including a seat cushion frame **100** capable of being folded up and down, a fixed frame assembly **200** configured to lock the seat cushion frame **100**, and a swivel device **400** connected to the seat frame assembly **10** to swivel the seat frame assembly **10**. Swiveling the seat frame assembly **10** may mean that the seat frame assembly **10** is rotated based on the position of the swivel device **400**. The seat frame assembly **10** may include a center frame **300**, a back frame **500**, and the seat cushion frame **100**. The back frame **500** and the seat cushion frame **100** may be coupled to the center frame **300**. The swivel device **400** may be connected to the lower portion of the center frame **300**. When the swivel device **400** is unlocked, the center frame **300** may be in a swivelable state.

[0048] The seat cushion frame **100** may be rotatably coupled to the center frame **300** to perform folded-up and folded-down functions using the center frame **300** as the center of rotation. Unlike the above-described example, the back frame **500** and the center frame **300** may be integrated, or the seat cushion frame **100** and the center frame **300** may be integrated.

[0049] The swivel device **400** connects the seat frame assembly **10** to the floor of a vehicle, and may rotate the seat frame assembly **10** connected to the floor of the vehicle. When rotating the seat frame assembly **10**, the seat frame assembly **10** may be rotated in the folded-up state of the seat cushion frame **100** considering the limited interior space of the vehicle.

[0050] However, if the seat cushion frame **100** is arbitrarily changed to the folded-down state by gravity or external force acting on the seat cushion frame **100** in the folded-up state, interference between the seat cushion frame **100** and a surrounding structure may occur when the seat frame assembly **10** is rotated. When the seat frame assembly **10** is rotated, the seat cushion frame **100** may be maintained in the folded-up state.

[0051] The fixed frame assembly **200** may unlock the swivel device **400**. The fixed frame assembly **200** may prevent the seat cushion frame **100** from being folded down if the seat frame assembly **10** is rotated by the swivel device **400**. Specifically, the fixed frame assembly **200** may maintain the seat cushion frame **100** in the folded-up state, but if the swivel device **400** is unlocked and the seat frame assembly **10** is rotated by a designated angle or more, when external force is applied to the seat cushion frame **100**, the seat cushion frame **100** may be folded down. The fixed frame assembly **200** may maintain the seat cushion frame **100** in the folded-up state while the swivel device **400** is locked or until the seat frame assembly **10** is not rotated by the designated angle or more in the unlocked state of the swivel device **400**.

[0052] FIG. 3 is a view showing the swivel device according to one embodiment of the present disclosure, and FIG. 4 is a view showing the swivel device located on a moving rail according to one embodiment of the present disclosure.

[0053] Referring to FIGS. 1, 3, and 4, the swivel device **400** may include a floor plate **410** located on a floor **910** of a vehicle, and a rotating plate **430** located on the floor plate **410**. Specifically, the floor plate **410** may be coupled to the floor **910** of the vehicle or a moving rail **930** provided on the

floor **910**. The swivel device **400** may be coupled to the moving rail **930** capable of moving the seat frame assembly **10** in the length direction of the vehicle or in the width direction of the vehicle, thereby being capable of varying a space where the seat frame assembly **10** is rotated in the interior space of the vehicle. The rotating plate **430** may be connected to the seat frame assembly **10** to rotate the seat frame assembly **10**. Specifically, the rotating plate **430** may be coupled to the lower surface of the center frame **300** of the seat frame assembly **10**. The rotating plate **430** may rotate the center frame **300** so that the entirety of the seat frame assembly **10** may be rotated.

[0054] The rotating plate **430** may be connected to the fixed frame assembly **200** through a wire. The swivel device **400** may be unlocked by operation of the fixed frame assembly **200**. When external force is applied to the seat frame assembly **10** in the unlocked state of the swivel device **400**, the seat frame assembly **10** may be swiveled.

[0055] FIG. **5** is a view showing the fixed frame assembly according to one embodiment of the present disclosure, and FIG. **6** is an exploded view of the fixed frame assembly of FIG. **5**.

[0056] Referring to FIGS. **2**, **5**, and **6**, the fixed frame assembly **200** may perform locking and unlocking of the swivel device **400** and the seat cushion frame **100**. The locking of the seat cushion frame **100** may mean fixing the seat cushion frame **100** so that the seat cushion frame **100** in the folded-up state is not folded down. The fixed frame assembly **200** may be disposed on the side surfaces of the seat frame assembly **10**. The side surfaces of the seat frame assembly **10** may mean both sides of the seat frame assembly **10** based on a direction in which the seat cushion frame **100** in the folded-down state extends.

[0057] The fixed frame assembly **200** may include a swivel lever **230** that performs the locking or unlocking of the swivel device **400**, a hook **250** that performs the locking or unlocking of the seat cushion frame **100**, and an elastic body **260** that applies elastic force to the swivel lever **230**. The swivel lever **230** and the hook **250** may be connected through a shaft **240**. Therefore, the swivel lever **230** and the hook **250** may be rotated together. The hook **250** may be connected to one end of the shaft **240**, and the swivel lever **230** may be connected to the other end of the shaft **240**.

[0058] The hook **250** may be disposed closer to the seat cushion frame **100**, compared to the swivel lever **230**. That is, the hook **250** and the swivel lever **230** may be sequentially disposed on the side surface of the seat cushion frame **100**.

[0059] The swivel lever **230** may be rotated about the shaft **240**. The shaft **240** may extend toward the side surface of the seat cushion frame **100**. That is, the shaft **240** may extend in a direction substantially perpendicular to the direction in which the seat cushion frame **100** in the folded-down state extends, and the swivel lever **230** may be rotated about the extending direction of the seat cushion frame **100** as an axis. For example, the swivel lever **230** may be fan-shaped. The swivel lever **230** may be connected to a wire **220**. The wire **220** may connect the swivel device **400** and the swivel lever **230**. As the swivel lever **230** is rotated, the swivel lever **230** may pull the wire **220** so as to unlock the swivel device **400**. Specifically, when the swivel lever **230** is rotated in a direction toward the front of the seat frame assembly **10**, the swivel lever **230** may pull the wire **220** so as to unlock the swivel device **400**. The direction toward the front of the seat frame assembly **10** may mean a direction in which the hook **250** is rotated toward a fixing pin **190**.

[0060] In order to unlock the swivel device **400**, the swivel lever **230** may apply external force to the swivel device **400**. The swivel lever **230** may be manually rotated by a user or automatically rotated by user's operation through a separate electric motor.

[0061] The hook **250** may be rotated together with the swivel lever **230** to fix the seat cushion frame **100**. The hook **250** may be rotated about the shaft **240**. The fixing pin **190** may be provided on the side surface of the seat cushion frame **100**. The fixing pin **190** may protrude toward the outside of the seat cushion frame **100** on the side surface of the seat cushion frame **100**. The hook **250** may be engaged with the fixing pin **190** by rotation. When the hook **250** is engaged with the fixing pin **190** in the folded-up state of the seat cushion frame **100**, the seat cushion frame **100** may be prevented from being folded down. However, when the hook **250** is rotated in the folded-down

state of the seat cushion frame **100**, the hook **250** may not be engaged with the fixing pin **190**. That is, the position of the fixing pin **190** may be changed when the seat cushion frame **100** is folded up or folded down.

[0062] The elastic body **260** may connect the seat cushion frame **100** and the swivel lever **230**. The elastic body **260** may be a torsion spring that receives a twisting force about a central axis. One end of the elastic body **260** may be connected to the seat cushion frame **100**, and the other end of the elastic body **260** may be connected to the swivel lever **230**. The central axis of the elastic body **260** may be located between the one end and the other end. The central axis of the elastic body **260** may coincide with the direction in which the fixing pin **190** protrudes toward the outside of the seat cushion frame **100**. That is, the central axis of the elastic body **260** may coincide with the protruding direction of the fixing pin **190**. One end of the elastic body **260** may be fixed to the seat cushion frame **100**, and the other end of the elastic body **260** may be connected to the swivel lever **230** so that the position of the other end of the elastic body **260** is variable. Therefore, the elastic body **260** may apply force in a direction from the one end to the other end thereof, and the direction of the force applied by the elastic body **260** to the swivel lever **230** may be varied depending on the position of the swivel lever **230**. The elastic body **260** may apply force to the swivel lever **230** so that the swivel lever **230** in the stopped state after completing movement maintains the position thereof. That is, the position of the elastic body **260** may be changed depending on rotation of the swivel lever **230**.

[0063] FIGS. **7** and **8** are views showing the locked state of the seat cushion frame according to one embodiment of the present disclosure.

[0064] Referring to FIGS. **2**, **7**, and **8**, the swivel lever **230** is operated in the folded-up state of the seat cushion frame **100**, thereby being capable of unlocking the swivel device **400**. The swivel lever **230** may be operated so that the hook **250** may be engaged with the fixing pin **190**. As the hook **250** is engaged with the fixing pin **190**, the seat cushion frame **100** may be prevented from being folded down. The elastic body **260** may apply force to the swivel lever **230** in a direction opposite to a direction in which the hook **250** moves away from the fixing pin **250**. In the state in which the hook **250** is engaged with the fixing pin **190**, the elastic body **260** may apply force to the swivel lever **230** in a direction A. The state in which the hook **250** is engaged with the fixing pin **190** may be maintained by the force applied by the elastic body **260** to the swivel lever **230**.

[0065] According to one embodiment of the present disclosure, if the swivel device **400** is unlocked in the folded-up state of the seat cushion frame **100**, the folded-up state of the seat cushion frame **100** may be maintained by the hook **250** rotated in conjunction with the swivel lever **230**, which is an element configured to unlock the swivel device **400**. Further, the engaged state between the hook **250** and the fixing pin **190** may be maintained by the force applied by the elastic body **260** to the swivel lever **230**.

[0066] FIG. **9** is a view showing the unlocked state of the seat cushion frame according to one embodiment of the present disclosure.

[0067] Referring to FIGS. **2** and **9**, the seat cushion frame **100** may be unlocked during a process of swiveling the seat frame assembly. If the seat cushion frame **100** is swiveled in the folded-up state of the seat cushion frame **100**, the hook **250** may be rotated from a state of being engaged with the fixing pin **190** to a state of being spaced apart from the fixing pin **190**. That is, if the swivel device **400** is unlocked and the seat frame assembly is rotated by the designated angle or more, the seat cushion frame **100** may be folded down when external force is applied to the seat cushion frame **100**.

[0068] In the process of swiveling the seat frame assembly, if the seat frame assembly is rotated by the designated angle or more, the swivel device **400** pulls the wire **220**. Accordingly, the swivel lever **230** may be rotated toward the rear of the seat frame assembly so that the hook **250** and the fixing pin **190** may be spaced apart from each other. The swivel lever **230** that has completed the rotation may receive force in a direction toward the rear of the seat frame assembly from the elastic

body **260**. The elastic body **260** may apply force to the swivel lever **230** in a direction B, i.e., the direction in which the hook **250** moves away from the fixing pin **190**. The swivel lever **230** may maintain the current position thereof by the force applied by the elastic body **260** to the swivel lever **230**. If the rotation of the hook **250** in the direction of moving away from the fixing pin **190** is completed, the swivel device **400** may be in the locked state.

[0069] FIG. **10** is a view showing one surface of the swivel device according to one embodiment of the present disclosure, and FIG. **11** is a view showing the other surface of the swivel device according to one embodiment of the present disclosure.

[0070] Referring to FIGS. **5**, **10**, and **11**, the swivel device **400** may include the floor plate **410** and the rotating plate **430**. The rotating plate **430** may be disposed in an internal space formed by the floor plate **410**.

[0071] The floor plate **410** may include a plurality of fixing grooves **411**. The fixing grooves **411** may be spaced apart from each other. For example, four fixing grooves **411** may be provided. Each of the fixing grooves **411** may mean a concave portion facing the outer surface of the floor plate **410**. In other words, the fixing grooves **411** may mean concave portions on the inner surface of the floor plate **410**. The inner surface of the floor plate **410** may face the outer surface of the rotating plate **430**.

[0072] The rotating plate **430** may include a rotating lever **431** connected to the wire **220**, locking pins **432** inserted into the fixing grooves **411**, an inner plate **433** that accommodates the locking pins **432**, and a lever rod **434** connected to the locking pins **432**. The locking pins **432**, the inner plate **433**, the lever rod **434**, and the rotating lever **431** may be sequentially disposed. A direction from the locking pins **432** toward the rotating lever **431** may be the direction from the swivel device **400** toward the seat frame assembly **10** of FIG. **1**.

[0073] The rotating lever **431** and the lever rod **434** may be fixedly connected to each other. An angle formed by the length direction of the rotating lever **431** and the length direction of the lever rod **434** may not be changed. One end of the rotating lever **431** may be connected to the wire **220**, and the other end of the rotating lever **431** may be connected to a central shaft **436** of the rotating plate **430**. A hole through which the central shaft **436** passes may be formed in the lever rod **434**.

[0074] The lever rod **434** may be connected to the inner plate **433** by springs **435**. The springs **435** may be connected to both ends of the lever rod **434**. The lever rod **434** may be connected to the locking pins **432** provided as a pair. As the position of the lever rod **434** is changed, the direction of force applied by the springs **435** to the lever rod **434** may be varied. For example, the springs **435** may be torsion springs.

[0075] The locking pins **432** may be disposed within locking pin receiving grooves **433a** provided in the inner plate **433**. For example, the locking pins **432** and the locking pin receiving groove **433a** may be provided as a pair. The locking pins **432** may be moved in the locking pin receiving grooves **433a**. Each of the locking pins **432** may include one end **432a** inserted into the fixing groove **411** and the other end **432b** opposite thereto. As the swivel lever **230** pulls the wire **220**, the rotating lever **431** and the lever rod **434** may be rotated, and accordingly, one end of each of the locking pins **432** connected to the lever rod **434** may be separated from the corresponding fixing groove **411**. That is, in the locked state of the swivel device **400**, one end **432a** of each of the locking pins **432** may be in a state of being inserted into one of the plurality of fixing grooves **411**. For example, if the locking pins **432** are provided as a pair, the respective ends **432a** of the pair of locking pins **432** may be inserted into two of the plurality of fixing grooves **411**. The pair of locking pins **432** may be disposed parallel to each other, and the direction in which one end **432a** of one locking pin **432** of the pair of locking pins **432** faces and the direction in which the other end **432b** of the other locking pin **432** faces may coincide with each other. The central shaft **436** may be located in a space between the pair of locking pins **432**, and the central shaft **436** may be connected to the inner plate **433**. The inner plate **433** may be disc-shaped, and may be rotated in the floor plate **410**. The inner plate **433** may be rotated by external force due to a swivel motion. That is, the

inner plate **433** may be connected to the seat frame assembly to be rotated by the rotation of the seat frame assembly.

[0076] FIG. **12** is a view for explaining the inner surface of the floor plate according to one embodiment of the present disclosure.

[0077] Referring to FIGS. **11** and **12**, the floor plate **410** may include the plurality of fixing grooves **411** and a plurality of guides **412**. The fixing grooves **411** may be the concave portions provided on the inner surface of the floor plate **410**. The guides **412** may be portions that protrude toward the inner plate **433** on the inner surface of the floor plate **410**. The fixing grooves **411** and the guides **412** may be formed at different elevations in the direction from the swivel device **400** toward the seat frame assembly. For example, the fixing grooves **411** and the guides **412** may be provided in the same number, and four fixing grooves **411** and four guides **412** may be provided.

[0078] One end **432a** of the locking pin **432** may be inserted into one of the fixing grooves **411**, and the other end **432b** of the locking pin **432** may come in contact with one of the guides **412**. The one end **432a** of the locking pin **432** and the other end **432b** of the locking pin **432** may be provided to be formed in layers with different elevations. The one end **432a** of the locking pin **432** and the other end **432b** of the locking pin **432** may form a stepwise shape so as to match the fixing grooves **411** and the guides **412** formed at different heights. That is, the heights of the one end **432a** of the locking pin **432** and the other end **432b** of the locking pin **432** may be different from each other based on one surface of the locking pin **432** facing in the direction from the swivel device **400** toward the seat frame assembly.

[0079] FIG. **13** is a view showing the folded-up state of the seat cushion frame according to one embodiment of the present disclosure, and FIGS. **14** and **15** are views the unlocked state of the swivel device according to one embodiment of the present disclosure.

[0080] Referring to FIGS. **8** and **13** to **15**, when the swivel lever **230** pulls the wires **220** in the folded-up state of the seat cushion frame **100**, the swivel device **400** may be unlocked. Specifically, when the wire **220** is pulled by the swivel lever **230**, the rotating lever **431** is rotated, and the locking pins **432** may be separated from the fixing grooves **411** by the rotation of the rotating lever **431** and the lever rod **434**. At this time, both the ends **432a** of the pair of locking pins **432** may be separated from the fixing grooves **411**. As the locking pins **432** are separated from the fixing grooves **411**, the seat frame assembly **10** may be in the swivelable state. The springs **435** may apply force in a direction in which the lever rod **434** is rotated, and accordingly, the locking pins **432** may be maintained in the state of being separated from the fixing grooves **411**. For example, the directions of forces applied by the two springs **435** to one end and the other end of the lever rod **434** may be parallel to each other.

[0081] FIG. **16** is a view showing the seat frame assembly according to one embodiment of the present disclosure that is swiveled. Referring to FIGS. **10**, **11**, and **16**, the seat frame assembly **10** may be swiveled in the unlocked state of the swivel device **400**. When the swivel motion is performed by applying external force to the seat frame assembly **10**, the inner plate **433** connected to the seat frame assembly **10** may be rotated. As the springs **435** applies force in the direction in which the lever rod **434** is rotated, the locking pins **432** may be maintained in the state of being separated from the fixing grooves **411**, and the inner plate **433**, the locking pins **432**, the rotating lever **431**, and the lever rod **434** may be rotated together.

[0082] FIGS. **17** and **18** are views showing a process of swiveling the swivel device according to one embodiment of the present disclosure.

[0083] Referring to FIGS. **5** and **16** to **18**, when the other end **432b** of each of the locking pins **432** comes in contact with one of the guides **412**, the locking pins **432** may be moved toward the inner surface of the floor plate **410**. When the locking pins **432** are moved toward the external part of the inner plate **433**, the rotating lever **431** and the lever rod **434** linked with the locking pins **432** may be rotated to pull the wire **220**. As the rotating lever **432** pulls the wire **220**, the swivel lever **230** may be rotated toward the rear of the seat frame assembly **10**, the hook **250** and the fixing pin **190**

may be spaced apart from each other. When the hook **250** and the fixing pin **190** do not contact each other, the locking of the seat cushion frame **100** is released, and thus the seat cushion frame **100** is capable of being folded down.

[0084] However, since the rotating lever **431** may be rotated together with the inner plate **433** even if the inner plate **433** is rotated by external force until the other end **432b** of each of the locking pins **432** comes in contact with one of the guides **412** after one end **432a** of each of the locking pins **432** is separated from the fixing groove **411**, the rotating lever **431** may not pull the wire **220**.

Therefore, until the other end **432b** of each of the locking pins **432** comes in contact with one of the guides **412** after one end **432a** of each of the locking pins **432** is separated from the fixing groove **411**, connection between the hook **250** and the fixing pin **190** may be maintained.

[0085] As the inner plate **433** is rotated by the rotation of the seat frame assembly **10**, the other ends **432a** of the locking pins **432** may come in contact with the guides **412**. As the inner plate **433** is continuously rotated in the contact state between the other ends **432a** of the locking pins **432** and the guides **412**, the locking pins **432** may be moved toward the external part of the inner plate **433**. That is, during the swivel process of the seat frame assembly **110**, the swivel lever **230** may start to be rotated at a point in time when the other ends **432a** of the locking pins **432** come in contact with the guides **412**, and thereby contact between the hook **250** and the fixing pin **190** may be eliminated. The locking of the seat cushion frame **100** is maintained at a point in time when the seat frame assembly **10** starts to be swiveled, but the seat cushion frame **100** may be unlocked when the rotation of the seat frame assembly **10** is performed at a designated angle.

[0086] FIGS. **19** and **20** are views showing a state in which swiveling of the swivel device according to one embodiment of the present disclosure is completed.

[0087] Referring to FIGS. **9**, **19**, and **20**, if the one end **432a** of each of the locking pins **432** is inserted into one of the fixing grooves **411** again, swiveling of the seat frame assembly **10** may be completed. During the process of inserting the ends **432a** of the locking pins **432** into the fixing grooves **411**, the rotating lever **431** and the lever rod **434** linked with the locking pins **432** may be rotated to completely pull the wire **220**. Accordingly, the swivel lever **230** may be rotated in the direction toward the rear of the seat frame assembly **10**. In the state in which the rotation of the swivel lever **230** is completed, the elastic body **260** may apply force to the swivel lever **230** so that the swivel lever **230** may maintain the current position thereof. The springs **435** may apply force to the lever rod **434** so that the locking pins **432** may be maintained in the state of being inserted into the fixing grooves **411**.

[0088] As is apparent from the above description, according to one embodiment of the present disclosure, if a swivel device is unlocked in a state in which a seat cushion frame is folded up, the folded-up state of the seat cushion frame may be maintained by a hook rotated in conjunction with a swivel lever configured to unlock the swivel device. In addition, engagement between the hook and a fixing pin connected to the seat cushion frame may be maintained by force applied by an elastic body to a swivel lever.

[0089] According to one embodiment of the present disclosure, the swivel lever configured to unlock the swivel device may be connected to the seat cushion frame by the elastic body, and the elastic body may apply force to the swivel lever to maintain the current state of the swivel lever, thereby being capable of maintaining the locked state and the unlocked state of the swivel device.

[0090] The disclosure has been described in detail with reference to preferred embodiments thereof. However, it will be appreciated by those skilled in the art that changes may be made in these embodiments without departing from the principles and spirit of the disclosure, the scope of which is defined in the appended claims and their equivalents.

Claims

- 1.** A vehicle seat comprising: a seat frame assembly comprising a seat cushion frame being foldable up and down; a swivel device configured to swivel the seat frame assembly; a swivel lever configured to unlock the swivel device; and a hook rotated together with the swivel lever to fix the seat cushion frame.
- 2.** The vehicle seat of claim 1, further comprising a fixing pin configured to protrude toward an outside of the seat cushion frame, wherein the hook is configured to be engaged with the fixing pin by rotation.
- 3.** The vehicle seat of claim 2, wherein, when the seat cushion frame is in a folded-up state and the swivel device is unlocked by the swivel lever, the hook is engaged with the fixing pin to prevent the seat cushion frame from being folded down.
- 4.** The vehicle seat of claim 3, further comprising a shaft configured to connect the hook and the swivel lever.
- 5.** The vehicle seat of claim 3, further comprising an elastic body configured to connect the swivel lever and the seat cushion frame, wherein the elastic body applies force to the swivel lever so that the hook is maintained in a state of being engaged with the fixing pin.
- 6.** The vehicle seat of claim 2, further comprising an elastic body configured to connect the swivel lever and the seat cushion frame, wherein, when the seat cushion frame is in a folded-up state and the swivel device is in a locked state in which no external force is applied to the swivel device, the elastic body applies force to the swivel lever such that the hook moves away from the fixing pin.
- 7.** The vehicle seat of claim 2, wherein, when the seat frame assembly is swiveled in a folded-up state of the seat cushion frame, the hook is configured to be rotated away from the fixing pin.
- 8.** The vehicle seat of claim 7, further comprising an elastic body configured to connect the swivel lever and the seat cushion frame, wherein, in a state where the rotation of the hook away from the fixing pin is completed, the elastic body is configured to apply force to the swivel lever such that the hook moves away from the fixing pin.
- 9.** The vehicle seat of claim 7, wherein the swivel device comprises: a floor plate coupled to a floor of a vehicle or a moving rail provided on the floor of the vehicle, the floor plate including fixing grooves; and a rotating plate provided on the floor plate and coupled to the seat frame assembly to allow the seat frame assembly to be swivelable based on external force, wherein the swivel lever is connected to the rotating plate of the swivel device through a wire.
- 10.** The vehicle seat of claim 9, wherein the rotating plate comprises: a rotating lever connected to the wire; an inner plate comprising locking pin receiving grooves configured to receive locking pins so that the locking pins are movable thereinto; and the locking pins configured to be moved in conjunction with the rotating lever and configured to be moved within the locking pin receiving grooves in a direction of being inserted into the fixing grooves, wherein, when the wire is pulled by the swivel lever, the rotating lever is configured to be rotated, and the locking pins are configured to be separated from the fixing grooves by the rotation of the rotating lever so that the seat frame assembly becomes swivelable.
- 11.** The vehicle seat of claim 10, wherein the fixing grooves of the floor plate are provided as concave portions on an inner surface of the floor plate, and a plurality of guides are formed to protrude toward the inner plate on the inner surface of the floor plate, and wherein the fixing grooves and the guides are formed at different elevations in a direction from the swivel device toward the seat frame assembly.
- 12.** The vehicle seat of claim 11, wherein one end of each of the locking pins is configured to be inserted into one of the fixing grooves, and a remaining end of each of the locking pins is configured to come in contact with one of the guides during a process of swiveling the seat frame assembly, wherein the one end and the remaining end of each of the locking pins are formed at different elevations in a stepwise manner.
- 13.** The vehicle seat of claim 12, wherein the one end of each of the locking pins is configured to

be separated from the one of the fixing grooves by the rotation of the swivel lever and the remaining end of each of the locking pins is configured to come in contact with the one of the guides during the process of swiveling the seat frame assembly, and wherein, when the remaining end of each of the locking pins comes in contact with the one of the guides, the one end of each of the locking pins is configured to be moved externally of the inner plate so that the wire is pulled by the rotating lever.

14. The vehicle seat of claim 13, wherein the wire is configured to be pulled by the rotating lever so that the hook is configured to move away from the fixing pin and the seat cushion assembly can be folded down.

15. The vehicle seat of claim 1, wherein the seat frame assembly further comprises a center frame and a back frame, and the back frame and the seat cushion frame are coupled to the center frame, and wherein the center frame and the swivel device are connected such that, when the swivel device is unlocked, the center frame becomes swivelable.

16. The vehicle seat of claim 1, further comprising a fixing pin configured to protrude toward an outside of the seat cushion frame, wherein, when the seat cushion frame is swiveled in a folded-up state of the seat cushion frame, the hook is configured to be rotated from a state of being engaged with the fixing pin to a state of being spaced apart from the fixing pin.
